

Week 1 · Task 2 – Networking Basics – IP, Ports, Protocols & OSI Model

1. IP Address vs. MAC Address

Network communication relies on two primary forms of identification: **MAC Addresses** and **IP Addresses**. While both serve to identify devices on a network, they operate at different layers of the OSI model and fulfill distinct roles in data delivery.

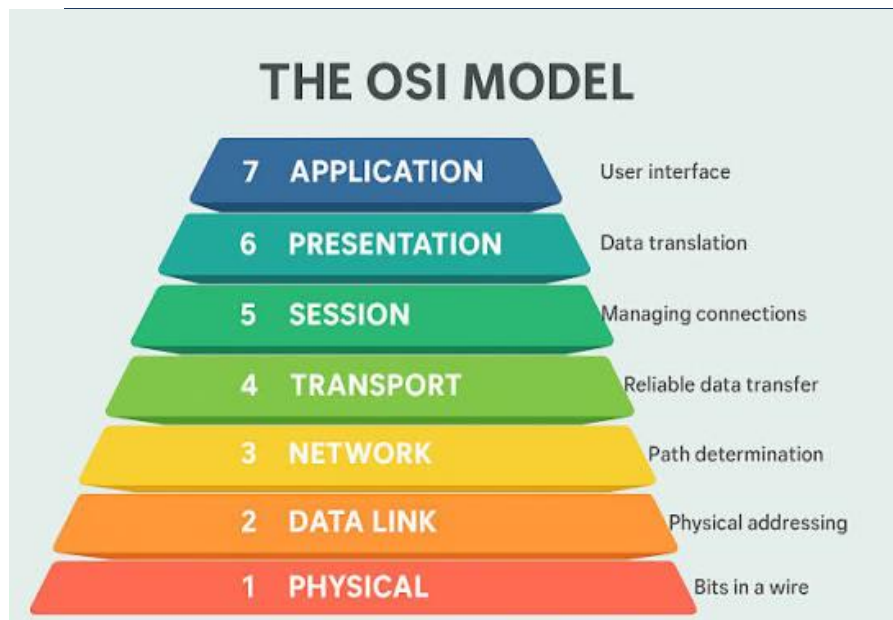
Feature	MAC Address (Media Access Control)	IP Address (Internet Protocol)
Layer & Role	Data Link Layer (Layer 2) Physical/Hardware address burned into NIC.	Network Layer (Layer 3) Logical/Software address assigned by network admin/DHCP.
Format & Size	48-bit (6 bytes) Hexadecimal Example: 00:1A:2C:3B:4D:5E	IPv4: 32-bit (4 bytes) Decimal (192.168.1.1) IPv6: 128-bit Hexadecimal (2001:db8::1)
Scope & Mobility	Local Area Network (LAN) communication only. Permanently tied to device hardware.	Global / Routing across networks (WAN/Internet). Dynamic and changes based on location.
Resolution Protocol	Address Resolution Protocol (ARP) translates IP to MAC for local delivery.	Domain Name System (DNS) translates domain names (e.g., google.com) to IP addresses.

2. OSI 7-Layer Reference Model

The Open Systems Interconnection (OSI) model standardizes network communication functions into seven distinct logical layers. Data travels down the stack (encapsulation) at the sender and up the stack (decapsulation) at the receiver.

Layer #	Layer Name	Protocol Data Unit (PDU)	Key Functions & Protocols
Layer 7	Application	Data	User interface, network services, application protocols. Protocols: HTTP, HTTPS, FTP, SMTP, DNS, SSH

Layer 6	Presentation	Data	Data formatting, encryption, compression, syntax translation. Formats: SSL/TLS, JPEG, ASCII, PNG, MPEG
Layer 5	Session	Data	Establishes, manages, and terminates sessions between applications. Protocols: NetBIOS, PPTP, RPC, Sockets
Layer 4	Transport	Segment (TCP) / Datagram (UDP)	End-to-end delivery, flow control, error recovery, port addressing. Protocols: TCP, UDP
Layer 3	Network	Packet	Logical addressing (IP) and path determination (routing). Protocols: IPv4, IPv6, ICMP, ARP, OSPF, BGP
Layer 2	Data Link	Frame	Physical addressing (MAC), error detection (CRC), framing. Protocols: Ethernet, Wi-Fi (802.11), PPP, Switches
Layer 1	Physical	Bit	Transmission of raw binary bits over physical medium. Hardware: Cables (Cat6, Fiber), Hubs, Repeaters, Signal standards



3. Common Network Ports & Protocols

Port numbers (0–65535) allow computer systems to specify which application service a network packet is destined for. Port numbers 0–1023 are Well-Known Ports.

Port Number	Protocol / Service	Transport	Description / Primary Usage
20 / 21	FTP (File Transfer Protocol)	TCP	Transferring files between client and server (Data / Command).
22	SSH / SFTP	TCP	Secure Shell for encrypted remote terminal access and file transfer.
23	Telnet	TCP	Unencrypted text communications for remote terminal access (Insecure).
25	SMTP (Simple Mail Transfer)	TCP	Sending and routing email between mail servers.
53	DNS (Domain Name System)	UDP / TCP	Resolves domain names to IP addresses (TCP used for zone transfers).
67 / 68	DHCP (Dynamic Host Config)	UDP	Dynamic IP address allocation (67=Server, 68=Client).
80	HTTP (Hypertext Transfer)	TCP	Unencrypted web browser traffic.
110	POP3 (Post Office Protocol)	TCP	Retrieving emails from server (downloads and typically deletes).
123	NTP (Network Time Protocol)	UDP	Clock synchronization across network devices.
143	IMAP (Internet Message Access)	TCP	Retrieving emails from server (syncs while leaving mail on server).
161 / 162	SNMP (Simple Net Mgmt)	UDP	Network device monitoring and metrics collection (161=Poll, 162=Trap).
389	LDAP (Lightweight Directory)	TCP / UDP	Directory services authentication (e.g., Active Directory).
443	HTTPS (HTTP Secure)	TCP	Encrypted web browser traffic over TLS/SSL.
445	SMB (Server Message Block)	TCP	Shared file and printer access in Windows networks.

3389	RDP (Remote Desktop)	TCP / UDP	Graphical remote desktop connection for Windows hosts.
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4. Transmission Protocols: TCP vs. UDP

At the Transport Layer (Layer 4), **TCP (Transmission Control Protocol)** and **UDP (User Datagram Protocol)** govern how data is packaged, transmitted, and verified across network boundaries.

Comparison Metric	TCP (Transmission Control Protocol)	UDP (User Datagram Protocol)
Connection Type	Connection-oriented (Requires explicit 3-way handshake SYN -> SYN-ACK -> ACK before sending data).	Connectionless (No handshake prior to data transmission; fire and forget).
Reliability	Guaranteed delivery. Lost packets are automatically retransmitted.	Unreliable / Best-effort delivery. Lost packets are dropped without notification.
Packet Ordering	Guaranteed order. Packets are sequenced and reassembled in original sequence.	No order guaranteed. Packets arrive out of order or independently.
Header Overhead	Heavy overhead (20–60 bytes) containing sequence numbers, flags, window size.	Lightweight overhead (8 bytes fixed size). Minimal field structure.
Flow & Congestion Control	Supports sliding window flow control and congestion control mechanisms.	No flow control or congestion management. Sends data as fast as application permits.
Speed & Performance	Slower due to handshake, acknowledgment overhead, and error recovery.	Extremely fast and lightweight with minimal latency.
Broadcasting Support	Unicast only (One-to-One communication).	Supports Unicast, Multicast, and Broadcast (One-to-Many / One-to-All).
Use Cases	Web browsing (HTTP/S), File Transfer (FTP), Email (SMTP/IMAP), Remote Access (SSH).	Live Streaming, Online Gaming, VoIP (Voice over IP), DNS queries, Video Conferencing.

Completed rooms
Created rooms
Skills matrix
Certificates
Badges
Yearly activity

DNS in Detail

Learn how DNS works and how it helps you access internet services.

Easy

Free Walkthrough

What is Networking?

Begin learning the fundamentals of computer networking in this bite-sized and interactive module.

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Intro to LAN

Learn about some of the technologies and designs that power private networks

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Pentesting Fundamentals

Learn the important ethics and methodologies behind every pentest.

Easy

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Security Principles

Learn about the security triad and common security models and principles.

Offensive Security Intro

Hack your first website (legally in a safe environment) and experience an ethical hacker's job.

Careers in Cyber

Discover the perks and your place within Cyber Security.

Task 1 Introduction

Task 2 OSI Model

Before we start, we should note that the OSI model might initially seem complicated. Don't worry if you encounter cryptic acronyms, as we provide examples of the OSI model layers. We assure you that by the time you finish this module, this task will feel like a piece of cake.

The OSI (Open Systems Interconnection) model is a conceptual model developed by the International Organization for Standardization (ISO) that describes how communications should occur in a computer network. In other words, the OSI model defines a framework for computer network communications. Although this model is theoretical, it is vital to learn and understand as it helps grasp networking concepts on a deeper level. The OSI model is composed of seven layers:

1. Physical Layer
2. Data Link Layer
3. Network Layer
4. Transport Layer
5. Session Layer
6. Presentation Layer
7. Application Layer

The numbering starts with the physical layer being layer 1, while the top layer, the application layer, is layer 7. To help you remember the layers from bottom to top, you can use a mnemonic such as "Please Do Not Throw Spinach Pizza Away." You can check the Internet for other easy-to-remember acronyms if this helps you memorise them. Remembering the OSI model layers with their layer numbers is important; otherwise, you will struggle to understand terms such as "layer 3 switch" or "layer 7 firewall."

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